EE6083 VLSI Testing

Quartz-4b7e - HW3

1. 3-1 Signature Analysis

Figure 1 和 Figure 2 為電路中各節點與 Verilog 宣告之對應，及實現 $1+X+X^4$ 之 LFSR。

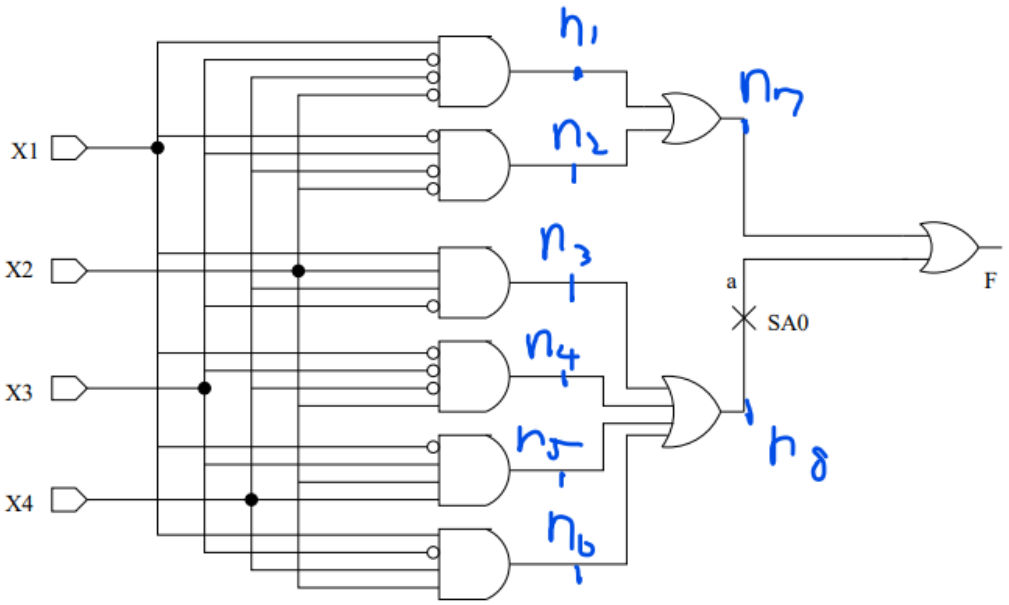


Figure 1 電路與節點宣告對應圖

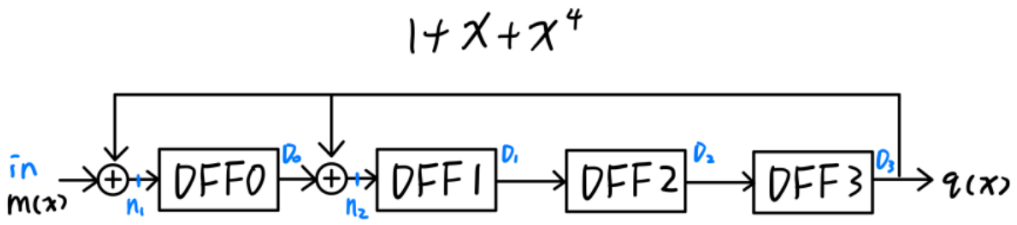


Figure 2 LFSR with characteristic function $1+X+X^4$

Figure 3 為 Figure 1 電路加上 Figure 2 之 LFSR 後之輸出波形圖，由 counter 從 0 數到 14，分別對應題目中的 15 個 input pattern，而經 LFSR 取出之 signature 則會在第 15 個 clock cycle (counter = 15)時在 signal D 得到。黃色線為 fault-free 之 output 與經 LFSR 取 signature 後的結果(a 小題)，綠色線為出現 SA0 fault 之 output 與 signature。

1. 從兩條黃色訊號線可以看到，當依序輸入 15 個 pattern，電路正確的 output 為 000010001000111 (signal *F1*)，而相對應之 signature 為 0011 (signal *D*): $d_0 = 0$ (DFF_0)、$d_1 = 0$ (DFF_1)、$d_2 = 1$ (DFF_2)、$d_3 = 1$ (DFF_3)
2. 從兩條綠色訊號線可以看到，當依序輸入 15 個 pattern，faulty output 為 000000000000101 (signal *F2*)，而相對應之 signature 為 0010(signal *D_faulty*):

$d_0 = 0$ (DFF_0)、$d_1 = 0$ (DFF_1)、$d_2 = 1$ (DFF_2)、$d_3 = 1$ (DFF_3)

從上面敘述中可以看到，當電路發生 single stuck-at fault 時，所輸出之 faulty output 經 LSFR 取 signature 後的結果(b)將會與 golden(a)不相同，因此能夠

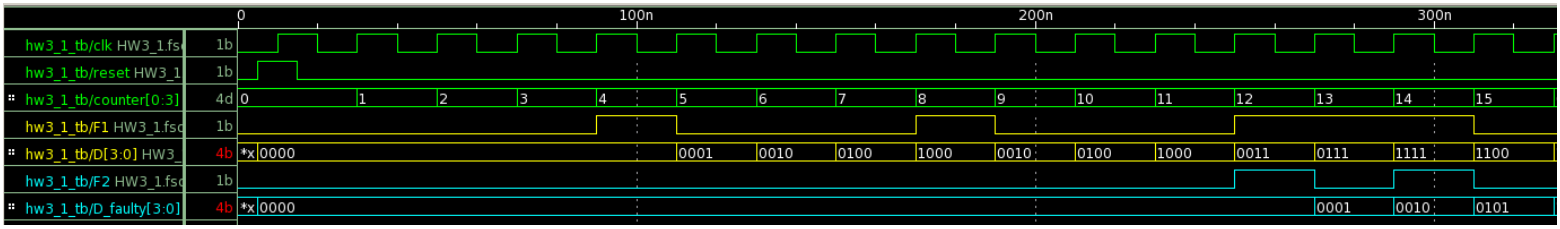


Figure 3 模擬波形圖

利用 LSFR 做 signature analysis 找出電路是否有 fault 發生。

3. 3-2 Syndrome Compaction

第二小題我透過 C++建立了一支電路模擬器，將 fault-free 時的 golden output 與透過 parity testing 及 one's count 所得到的 golden signature 存下來。再透過模擬各節點在發生 SA0、SA1 時的 faulty output 和所對應之 parity 與 one's count，與 golden 的結果做比較，確認哪個 fault 無法透過 parity testing 與 one's count 測出，以下為模擬器執行結果：

```
Fault free
Output: 00101010
Odd parity.
One's count: 3

SA0 at a
Output: 10101010
Even parity.
One's count: 4
Find fault by parity testing!
Find fault by one's count!

SA1 at a
Output: 00100010
Even parity.
One's count: 2
Find fault by parity testing!
Find fault by one's count!

SA0 at b
Output: 10101010
Even parity.
One's count: 4
Find fault by parity testing!
Find fault by one's count!

SA1 at b
Output: 00001010
Even parity.
One's count: 2
Find fault by parity testing!
Find fault by one's count!

SA0 at c
Output: 00000000
Even parity.
One's count: 0
Find fault by parity testing!
Find fault by one's count!

SA1 at c
Output: 00101110
Even parity.
One's count: 4
Find fault by parity testing!
Find fault by one's count!

SA0 at d
Output: 00000000
Even parity.
One's count: 0
Find fault by parity testing!
```

```
SA1 at d
Output: 00101010
Odd parity.
One's count: 3
Parity testing doesn't find any fault.
One's count doens't find any fault.

SA0 at e    Undetected fault
Output: 10101010
Even parity.
One's count: 4
Find fault by parity testing!
Find fault by one's count!

SA1 at e
Output: 00000000
Even parity.
One's count: 0
Find fault by parity testing!
Find fault by one's count!

SA0 at f
Output: 00111111
Even parity.
One's count: 6
Find fault by parity testing!
Find fault by one's count!

SA1 at f
Output: 00000000
Even parity.
One's count: 0
Find fault by parity testing!
Find fault by one's count!

SA0 at z
Output: 00000000
Even parity.
One's count: 0
Find fault by parity testing!
Find fault by one's count!

SA1 at z
Output: 11111111
Even parity.
One's count: 8
Find fault by parity testing!
Find fault by one's count!
```

Figure 4 模擬器執行結果

從上圖(Figure 4)中可以看到，在 fault-free 情況下，電路的輸出為 00101010，此時的 parity 為 odd parity、one's count 為 3。在發生 d/1 fault 時(Figure 4 紅色框框處)，電路 output 與 fault-free 時相同，經 parity test 與 one's count 計算後之結果也與 golden 的結果相同，因此不論是 parity test 還是 one's count 都無法測到這個 fault，因為出現 1 的奇偶數與 golden 相同，1 的數量也與 golden 相同。

而在其他 fault 發生的情況下，電路的 output 經 parity test 後皆為 even parity，因此都能夠過 parity test 測到; 同理 one's count 也是，從上圖中可以看到除了 d/1 外其他 fault 發生時的 output 經 one's count 計算後的結果都與 golden 不相同，代表都能夠被測到。

4. 附檔

1. circuit.v：第一小題題目給定的電路的 Verilog code
2. LFSR.v：第一小題產生 signature 所需之 LFSR 電路 Verilog code
3. HW3_1.v：將 circuit.v 與 LFSR.v 包成的 main circuit
4. HW3_1_tb.v：執行第一小題電路之 testbench
5. HW3_2.cpp：第二小題電路模擬器，實現 faulty 與 fault-free 行為與計算相對應之 parity 與 one's count